

A
PROJECT REPORT
ON
"STUDENT ASSIGNMENT MANAGEMENT SYSTEM "

A full-stack web-based application developed to digitize and automate assignment submission and evaluation in academic institutions.

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)
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[2024-2028]

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CERTIFICATE

This is to certify that this Real Time Research Project Report entitled "STUDENT ASSIGNMENT MANAGEMENT SYSTEM" is a Bonafide work carried out by Rangiseti Sreekar (24N81A6235), T Kiran Teja (24N81A6209), G Prahalad Reddy (24N81A6212), Yash Kumar Lidhorya (24N81A6233), students of B.Tech Computer Science and Engineering (Cyber Security), Sphoorthy Engineering College, during the academic year 2024-2028.

The work presented in this report describes a centralized digital platform where students upload assignments, PPTs, and Project-Based Learning (PBL) files under specific subjects. Teachers can securely access uploaded files, review submissions, assign marks, and download reports.

The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We the undersigned declare that the Real Time Research Project Report "Student Assignment Management System " carried out at "Sphoorthy Engineering College" is original and is being submitted to the Department of Computer Science and Engineering (Cyber Security) as part of the B.Tech RTRP/PBL work.

We declare that the result embodied in this Real Time Research Project Report has not been submitted to any other University or Institute for the award of any Degree or Diploma.

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BATCH: 2024 - 2028

ABSTRACT

The Student Assignment Management System is a full-stack web-based application developed to digitize, automate, and simplify the assignment submission and evaluation process in academic institutions. Traditional assignment management methods often rely on physical submissions, email sharing, and manual record maintenance, which can lead to inefficiency, data loss, delayed evaluations, and lack of transparency between students and teachers. Managing multiple assignments, projects, and student records manually becomes increasingly difficult as the number of students and subjects increases.

The proposed system provides a centralized and user-friendly platform where students can securely upload academic files such as Assignments, PowerPoint presentations (PPT), and Project Based Learning (PBL) documents for specific subjects. Instead of using multiple communication methods, all academic submissions are maintained within a single digital environment. This reduces confusion and ensures organized storage of data.

The application follows a role-based architecture consisting of separate dashboards for students and teachers. Students can log in using their credentials, select subjects from the available list, and upload required documents. Teachers can access uploaded files, review student submissions, assign marks, and monitor academic performance through an interactive dashboard. The system also supports viewing uploaded files directly from the teacher interface and downloading reports whenever required.

To improve data management and accessibility, uploaded files are securely stored and linked with student records in the database. The use of a structured database system helps maintain consistency and allows fast retrieval of information. Teachers can efficiently track submissions and update marks without relying on traditional paperwork.

The system is designed using modern web technologies including Spring Boot for backend development, HTML, CSS, and JavaScript for frontend implementation, MySQL for database management, and REST APIs for communication between client and server components. The use of these technologies ensures scalability, maintainability, and efficient performance.

The Student Assignment Management System minimizes manual effort, reduces paperwork, improves communication between students and teachers, and creates a transparent and efficient digital academic environment. The proposed system serves as a reliable solution for educational institutions seeking to modernize assignment handling and evaluation processes.

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CHAPTER-1

INTRODUCTION

PROBLEM STATEMENT

Traditional and semi-digital assignment submission systems often face problems such as manual record maintenance, scattered file sharing through emails or messaging platforms, delayed evaluation processes, difficulty in tracking submissions, and lack of centralized management. In many educational institutions, students submit assignments through multiple platforms, making it difficult for teachers to organize files and evaluate submissions efficiently. During peak academic periods, teachers may receive a large number of assignments, resulting in confusion and improper management.

The Student Assignment Management System is developed to solve this problem by providing a centralized digital platform for assignment submission and evaluation. The system enables students to upload assignments, PowerPoint presentations (PPT), and Project-Based Learning (PBL) files while allowing teachers to access submissions, assign marks, and monitor student activities through a dedicated dashboard.

OBJECTIVE

1. To develop a web-based assignment management system for students and teachers.
 2. To provide subject-wise assignment, PPT, and PBL file submission.
 3. To allow teachers to view uploaded files through a centralized dashboard.
 4. To enable marks allocation and management for each student.
 5. To reduce paperwork and manual assignment handling.
 6. To generate downloadable reports and academic records.
 7. To maintain student details, subjects, uploaded files, and marks using MySQL database.
-

EXISTING SYSTEM

In many existing educational systems, assignment submissions are managed through emails, messaging applications, or manual submissions. Teachers often need to download and organize files manually. During busy academic schedules, tracking student submissions becomes difficult and time-consuming. Some systems also require students to use multiple platforms for different subjects.

Drawbacks:

- Assignment records are difficult to manage manually.
- File organization becomes difficult for teachers.

- Students may not know submission status clearly.
 - Marks maintenance requires manual effort.
 - Different platforms create confusion.
 - Paper-based methods consume time and resources.
-

PROPOSED SYSTEM

The proposed Student Assignment Management System is a complete online academic platform that enables students to select subjects and upload assignments, PPTs, and PBL files digitally. Teachers can access uploaded documents through a dashboard, assign marks, download reports, and manage submissions efficiently.

Key Features of Proposed System

- Student login and authentication system.
- Subject-wise file upload support.
- Assignment, PPT, and PBL upload functionality.
- Teacher dashboard for viewing submissions.
- Marks allocation and management system.
- Downloadable reports and Excel generation.
- Centralized database storage.
- User-friendly interface for students and teachers.
- Secure file management system.

Working of the System

1. Student logs into the dashboard.
2. Student selects a subject from available subjects.
3. Student uploads Assignment, PPT, or PBL file.
4. Uploaded files are stored in the server uploads folder.
5. Teacher accesses the teacher dashboard.
6. Teacher selects a subject and views student files.
7. Teacher assigns marks and saves academic records.

SCOPE

The scope of this project includes the design and implementation of a web-based academic assignment management system. It covers student login, file upload management, teacher dashboard operations, marks management, report generation, and database handling.

Applications:

- College assignment submission systems.
- School academic portals.
- Project submission systems.
- Internal assessment management systems.
- Educational institutions with digital learning platforms.

Future Scope:

- Email notifications for assignment deadlines.
- AI-based plagiarism detection.
- Mobile application support.
- Cloud-based file storage.
- Analytics dashboard for performance monitoring.
- Push notifications and reminders.

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

- Operating System: Windows 10 / Windows 11
- Frontend: HTML, CSS, JavaScript
- Backend: Spring Boot
- Database: MySQL
- Build Tool: Maven
- IDE: Visual Studio Code
- APIs: REST APIs

Hardware Requirements

- RAM: 4 GB minimum, 8 GB recommended
- Storage: 50 GB free disk space minimum
- Processor: Intel Core i3 / Core i5 or equivalent AMD processor
- System Type: 64-bit operating system
- Network: Stable internet connection for testing and file uploads

CHAPTER-2

LITERATURE SURVEY

2.1 SURVEY OF MAJOR AREA RELEVANT TO PROJECT

Assignment Management Systems:

Assignment management systems allow students to upload academic files and teachers to review, evaluate, and maintain records digitally. These systems reduce paperwork and improve submission tracking efficiency.

Web-Based Educational Platforms:

Web-based educational platforms provide centralized access to academic resources, assignments, learning materials, and student records. Such systems improve accessibility and reduce dependency on physical document handling.

File Upload and Storage Systems:

Modern web applications use file upload systems to allow users to store documents securely. File handling mechanisms support different file formats and maintain uploaded records for future access and evaluation.

Database Management Systems:

Educational systems require reliable databases to store student information, subject details, uploaded files, assignment records, and marks. Database constraints and relationships help maintain data consistency and avoid duplication.

Authentication and Role-Based Access:

Secure login systems are important because students and teachers perform different tasks. Authentication mechanisms ensure that only authorized users access specific dashboards and system features.

Digital Evaluation Systems:

Digital evaluation systems allow teachers to review assignments, assign marks, and maintain academic records electronically. These systems reduce manual effort and improve academic transparency.

2.2 TECHNIQUES AND ALGORITHMS

Student Authentication

The system verifies student login credentials before providing dashboard access. Authentication mechanisms ensure that only registered users can upload assignments and access resources.

Subject-Based Assignment Management

Students select subjects from predefined lists. Uploaded files are organized subject-wise, making retrieval and management easier for teachers.

File Upload Algorithm

When a student uploads a file, the system validates the request, stores the file in the uploads directory, and saves file information in the database for future access.

Database Record Management

The system stores uploaded file details, student records, subjects, and marks using structured database tables with relationships and constraints.

Marks Allocation Technique

Teachers can access uploaded files and assign marks through the dashboard. Marks are stored and updated dynamically within the database.

Role-Based Routing

Access control mechanisms ensure that students can upload assignments while teachers can view files, assign marks, and generate reports.

2.3 APPLICATIONS

- College assignment submission systems.
- School academic management platforms.
- Internal assessment management systems.
- Project and seminar submission portals.
- Digital educational record systems.
- Teacher evaluation and report generation systems.
- Online student file management systems.

CHAPTER-3

SYSTEM DESIGN

System design describes the structure, workflow, modules, and interaction between users and system components of the Student Assignment Management System. The design focuses on secure file uploading, subject-wise assignment handling, teacher evaluation, role-based access, and efficient database management.

Modules of System Design

- Student Authentication Module
- Subject Management Module
- Assignment Upload Module
- File Storage and Retrieval Module
- Marks Evaluation Module
- Teacher Dashboard Module
- Student Dashboard Module
- Database Management Module
- Report Generation Module

3.1 SYSTEM ARCHITECTURE

The system follows a layered web application architecture. The presentation layer contains HTML, CSS, JavaScript, and Bootstrap-based interfaces for student and teacher dashboards. The application layer is developed using Spring Boot controllers, services, repositories, and REST APIs. The data layer uses MySQL database tables for students, subjects, uploaded assignments, and marks records.

The assignment upload mechanism works between the application layer and storage layer. When a student uploads a file, the server validates the request, stores the uploaded file in the uploads directory, and saves file details in the database. Teachers can retrieve these records and assign marks through the dashboard.

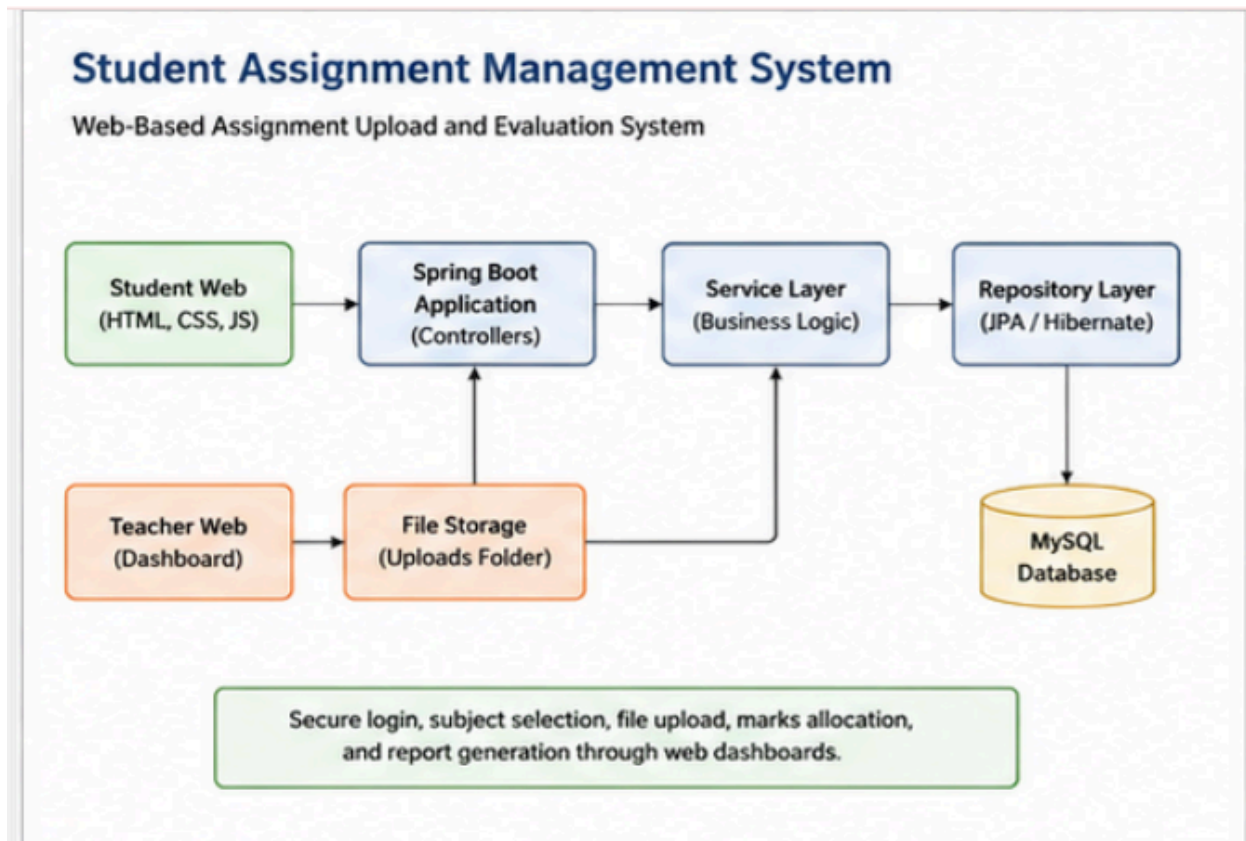


Figure 3.1: System Architecture Diagram

3.2 SYSTEM FLOW

The workflow begins when a student logs into the system and selects a subject. The student uploads assignment files and the system stores records. Teachers can later access uploaded files and assign marks.

1. Student logs into the system using credentials.
2. System validates login information.
3. Student dashboard displays available subjects.
4. Student selects a subject.
5. Student uploads assignment, PPT, or PBL file.
6. System stores file inside uploads folder.
7. Database stores uploaded file information.
8. Teacher accesses teacher dashboard.
9. Teacher selects subject and views uploaded files.
10. Teacher assigns marks and updates records.

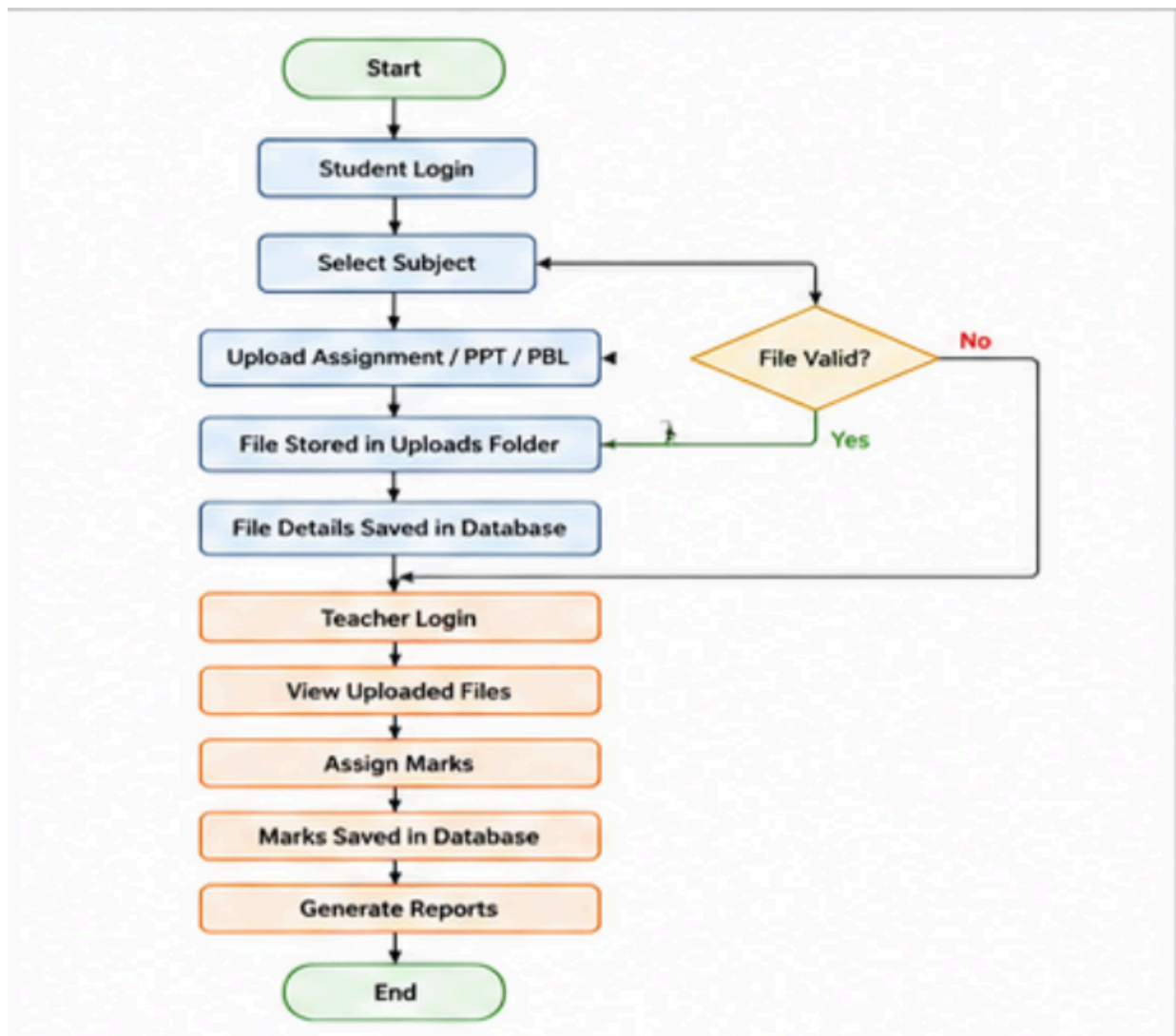


Figure 3.2: System Flow Diagram

3.3 MODULE DESCRIPTION

Student Authentication Module

Handles student login verification and dashboard access.

Subject Management Module

Displays predefined subjects and allows subject selection.

Assignment Upload Module

Allows students to upload assignment files, presentations, and project documents.

File Storage Module

Stores uploaded documents inside the upload directory and maintains database references.

Teacher Dashboard Module

Allows teachers to access subject-wise uploaded files and student records.

Marks Evaluation Module

Allows teachers to enter marks and save records into the database.

Student Dashboard Module

Displays available subjects and upload options for students.

Report Generation Module

Provides functionality for exporting records and generating academic reports.

Database Module

Manages relational tables, constraints, and database operations required for student assignment management.

3.4 UML DIAGRAM

The following UML diagrams are represented as placeholders and can be replaced with final diagrams during project submission.

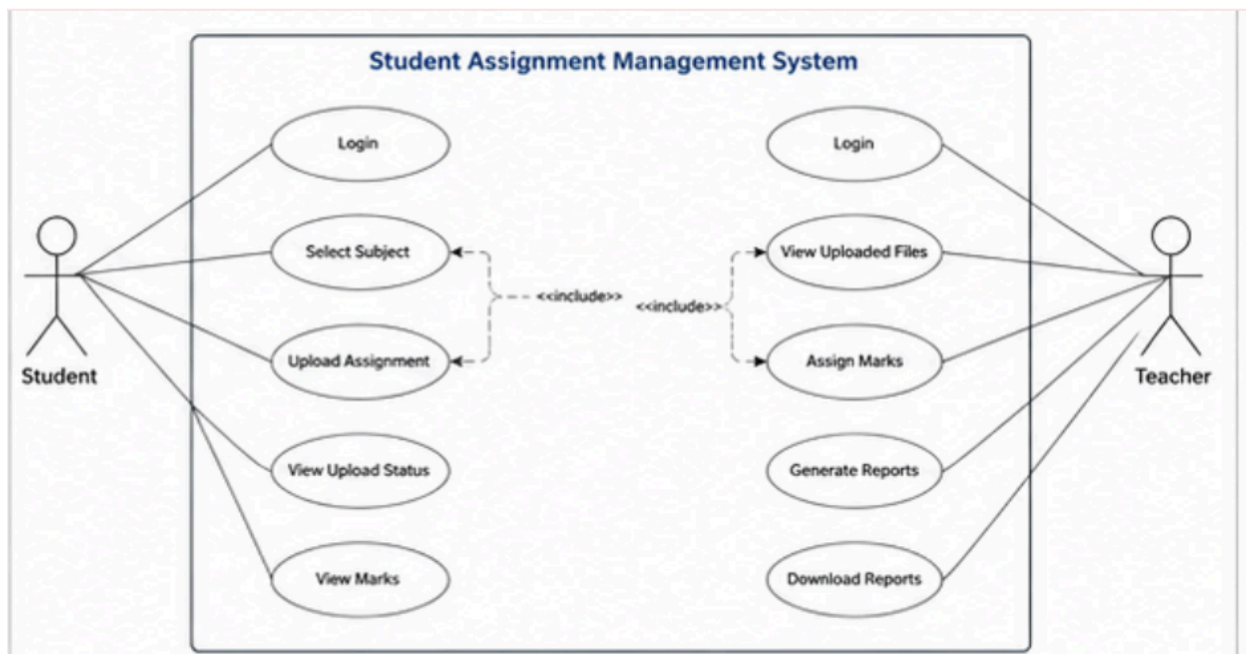


Figure 3.3: Use Case Diagram

Use cases: Login, Select Subject, Upload Assignment, View Files, Assign Marks, Generate Reports

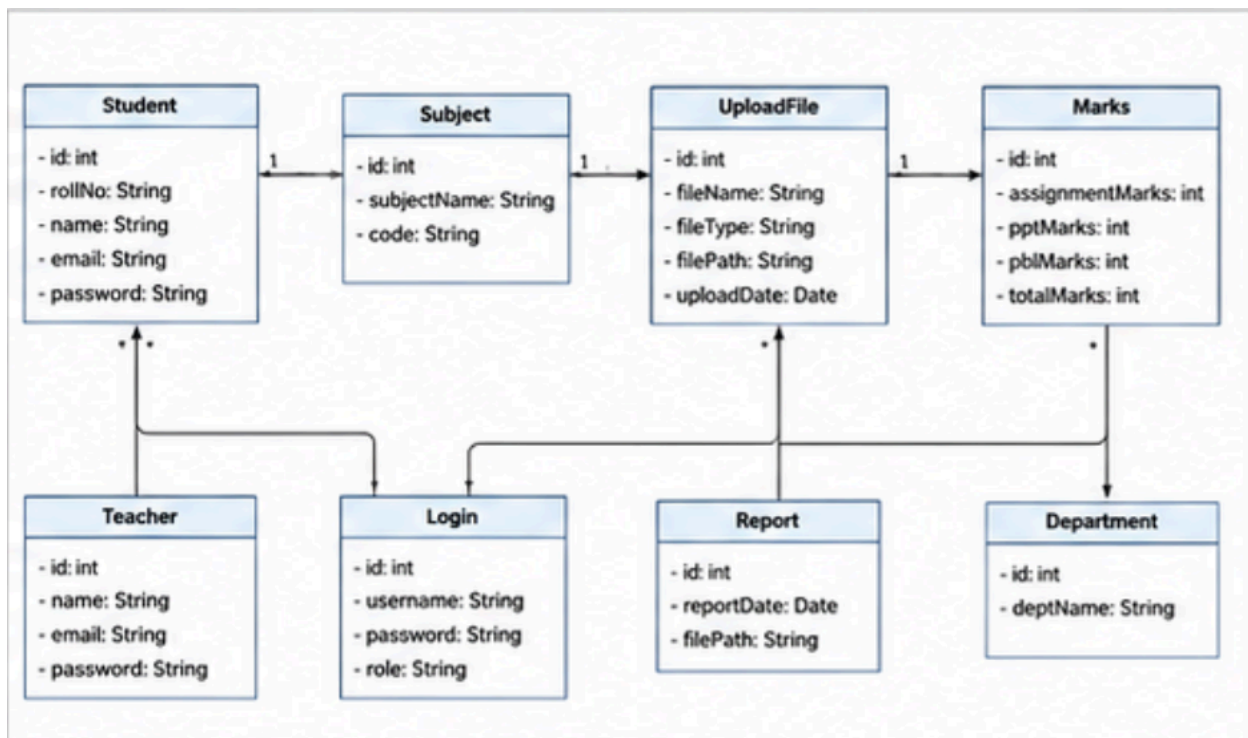


Figure 3.4: Class Diagram

Classes: Student, Subject, UploadFile, Marks, Teacher, Dashboard

Sequence:

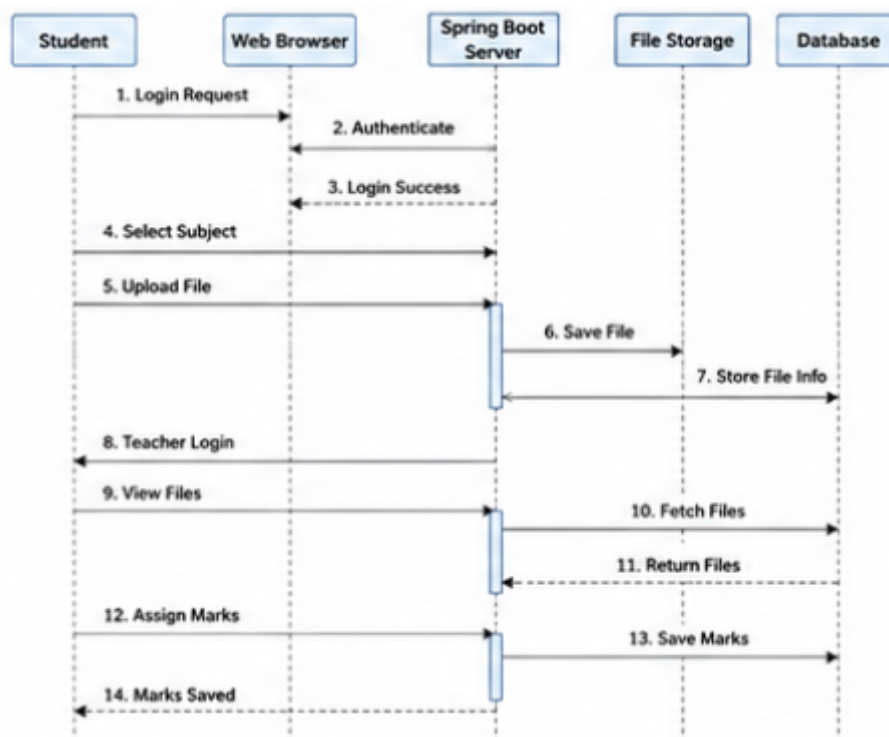


Figure 3.5: Sequence Diagram

Sequence: Student login → Subject selection → Upload file → Store file → Database update → Teacher views → Teacher assigns marks

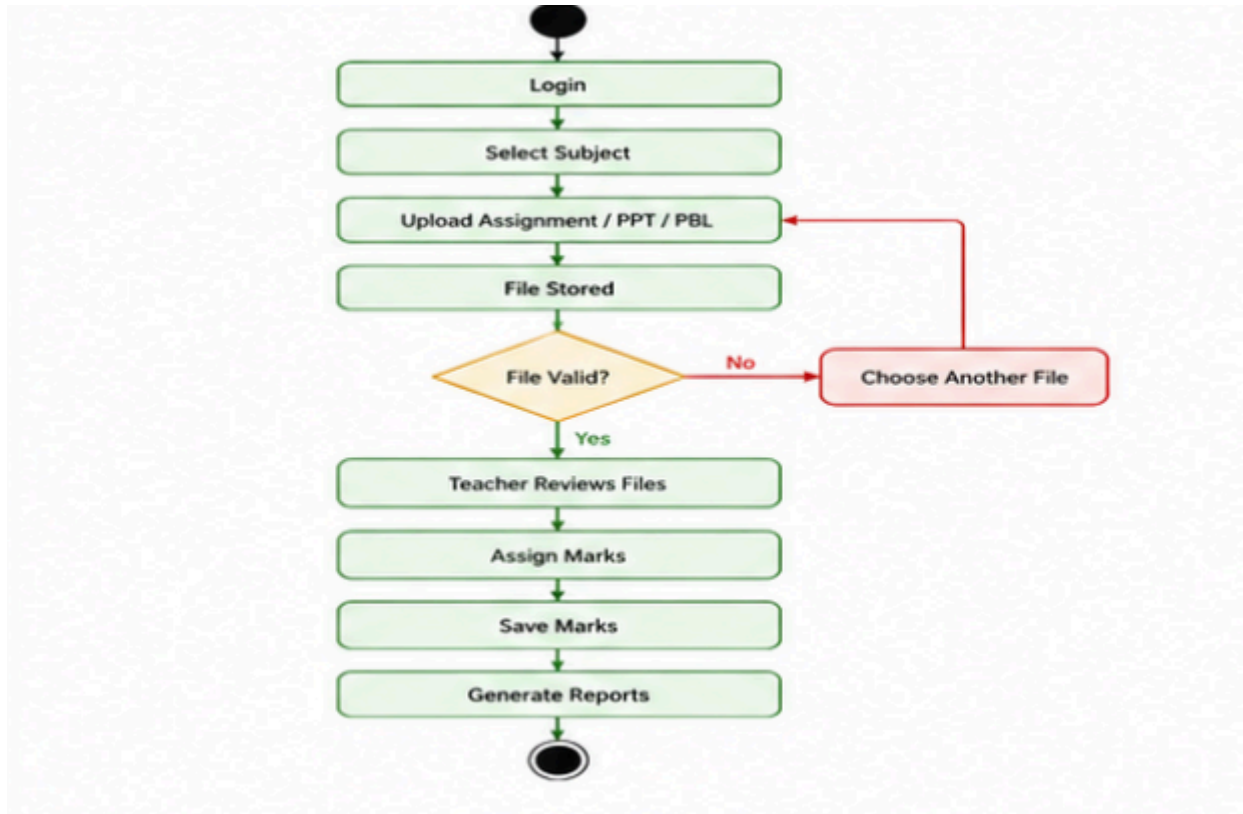


Figure 3.6: Activity Diagram

Activity: Login → Select Subject → Upload File → Save File → Teacher Review → Assign Marks → Generate Report

CHAPTER-4

IMPLEMENTATION

4.1 ENVIRONMENT SETUP

The Student Assignment Management System is implemented as a Spring Boot web application. Spring Boot manages REST APIs and backend processing, HTML/CSS/JavaScript provides the user interface, MySQL stores relational data, and file handling mechanisms manage assignment uploads and retrieval.

Component	Technology Used
Frontend	HTML, CSS, JavaScript
Backend	Spring Boot, Java
Database	MySQL
API Communication	REST APIs
Authentication	Student login validation
File Handling	Multipart file upload
Development Tool	Visual Studio Code

4.2 IMPLEMENTATION OF MODULES

Student Module

Students can log in to the system, select subjects, upload assignments, PPTs, and PBL files, and view submission status.

Subject Module

Subjects are displayed through predefined subject lists fetched from the backend API. Students can select a subject before submission.

File Upload Module

Students upload files through the dashboard interface. The system accepts uploaded files and stores them in the uploads directory while maintaining references inside the database.

Assignment Management Module

When a file is uploaded, the browser sends the file data to the backend API. The server validates request details, stores the file in the uploads folder, and updates the database record with file information.

Teacher Module

Teachers can access uploaded assignments, select subjects, view student submissions, and provide marks.

Marks Evaluation Module

Teachers can assign marks for uploaded assignments. Marks entered by teachers are saved and updated in the database.

Report Generation Module

Teachers can generate reports and download Excel sheets containing student records and marks.

Database Module

The database schema includes students, subjects, uploaded files, marks records, and file information required for assignment management.

4.3 INTEGRATION AND DEVELOPMENT

The system integrates Spring Boot controllers, repositories, HTML pages, JavaScript functions, and MySQL queries. Controllers handle API requests, repositories manage database operations, and frontend pages communicate through REST APIs.

The integration process begins when a student selects a subject and uploads a file. The request reaches the upload controller, which stores the file and updates database records. Teachers later access uploaded files through the dashboard and assign marks.

File Viewing Workflow Explanation

The current implementation stores uploaded files inside the uploads directory and generates file paths in the database. The application retrieves these paths while displaying uploaded files. When a teacher clicks a file link, the request accesses the uploaded file location and opens the document for viewing.

The file system integration ensures that uploaded documents remain associated with student records and subject information. This provides organized storage and efficient retrieval of assignment files.

CHAPTER-5

EVALUATION

5.1 MODULES

- Student Login Module
- Subject Selection Module
- Assignment Upload Module
- PPT Upload Module
- PBL Upload Module
- File Storage and Retrieval Module
- Teacher Dashboard Module
- Marks Evaluation Module
- Excel Report Generation Module
- Database Management Module

5.2 EVALUATION METRICS

Metric	Description
Functional Accuracy	Checks whether login, subject selection, file upload, and marks allocation work correctly.
File Upload Validation	Checks whether assignments, PPTs, and PBL files are uploaded successfully.
Response Time	Measures whether file upload and retrieval operations are performed quickly.
Database Integrity	Checks whether student records, files, subjects, and marks are stored correctly.
Usability	Checks whether students and teachers can perform operations easily.
Security	Checks login validation and unauthorized access prevention.
Scalability	Checks whether the system can support multiple students and subjects.

5.3 TEST CASES

Test Case	Module	Input/Action	Expected Result	Status
TC-01	Student Login	Enter valid student credentials	Student dashboard opens successfully	Pass
TC-02	Invalid Login	Enter incorrect login details	System displays validation error	Pass
TC-03	Subject Selection	Select a subject from dashboard	Selected subject page opens	Pass
TC-04	Assignment Upload	Upload assignment file	File uploads successfully	Pass
TC-05	PPT Upload	Upload PPT file	PPT file stored successfully	Pass

TC-06	PBL Upload	Upload PBL file	PBL file uploaded successfully	Pass
TC-07	File Validation	Upload unsupported file	System displays error message	Pass
TC-08	Teacher Dashboard	Teacher opens dashboard	Uploaded student records are displayed	Pass
TC-09	View Uploaded Files	Teacher clicks file view option	Uploaded file opens successfully	Pass
TC-10	Marks Assignment	Teacher enters marks and saves	Marks are stored in database	Pass
TC-11	Excel Report	Click download Excel option	Excel report downloads successfully	Pass
TC-12	Database Record Check	Verify uploaded file details	Records stored correctly in database	Pass

Requirement Analysis

The system requirements were analyzed from the need for an assignment management platform that allows students to upload files and teachers to evaluate submissions efficiently.

Test Planning

Testing was planned around login functionality, subject selection, file upload operations, teacher dashboard access, marks assignment, and database consistency.

Test Case Design

Test cases were designed to cover normal student uploads, invalid file submissions, subject selection errors, teacher evaluation actions, and report generation operations.

Test Environment Setup

The test environment includes Spring Boot, MySQL database, Visual Studio Code, browser clients, and sample student and teacher records.

Bug Reporting

Any issue identified during testing is recorded with module name, input condition, expected result, actual result, and correction status.

Bug Fixing and Retesting

After fixing issues, the same test cases are executed again to verify that the expected output is obtained without affecting other modules.

Final Testing

Final testing confirms that students can upload files successfully and teachers can access submissions and assign marks properly.

Test Closure

Testing is completed after all important modules perform correctly and no major file upload or database issues remain.

5.4 RESULTS

Results Obtained

- The system successfully displays subject information to students.
- Students can upload assignments, PPT files, and PBL files successfully.
- Uploaded files are stored and linked with student records.
- Teachers can view uploaded submissions through the dashboard.
- Marks can be assigned and saved into the database successfully.
- Excel reports can be generated for academic records.
- Student and teacher modules operate correctly through REST API integration.
- Database records remain consistent and accessible throughout the workflow.

Welcome Back!

Username

Password

LOGIN

Figure 5.1: Home/Landing Page

Student Assignment Management System

Dashboard

Subjects

Upload Assignment

My Assignments

Profile

Logout

Dashboard

Welcome back, Student!

5
Subjects

12
Assignments
Uploaded

8
PPTs
Uploaded

6
PBLs
Uploaded

Recent Uploads

Subject	Type	File Name	Date
Java Programming	Assignment	Java_Notes.pdf	20 May 2025
Data Structures	PPT	DS_Presentation.pptx	19 May 2025
Web Development	PBL	Web_Project.zip	18 May 2025

View All

Figure 5.2: Student Dashboard

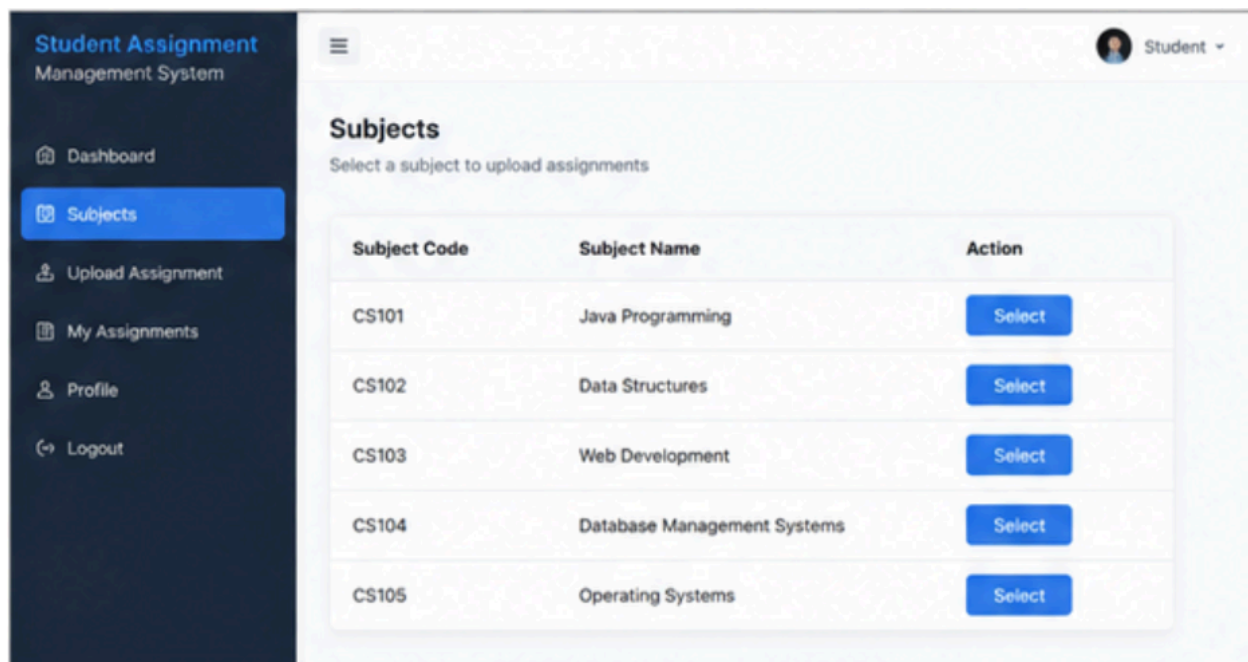


Figure 5.3:Subject Landing Page

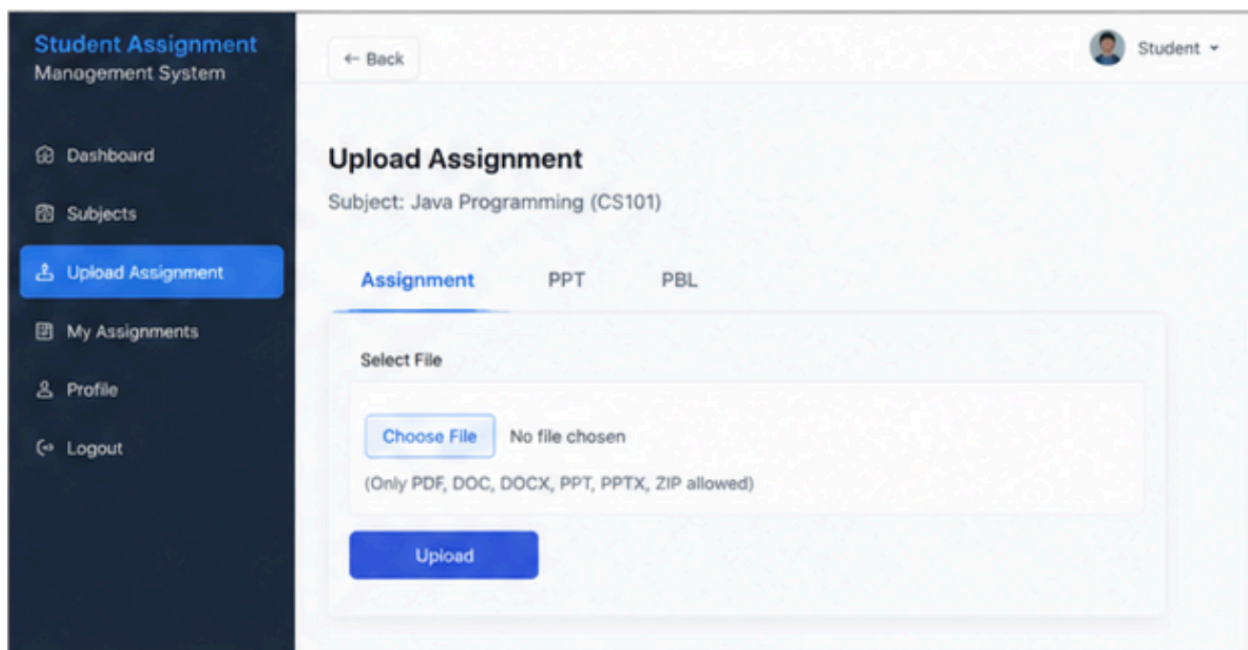


Figure 5.4:File Upload Page(Assignmnet)

Student Assignment Management System

Dashboard
Subjects
Upload Assignment
My Assignments
Profile
Logout

Student

My Assignments

View all your uploaded assignments

Subject	Type	File Name	Upload Date	Status	Action
Java Programming	Assignment	Java_Notes.pdf	20 May 2025	Uploaded	View
Data Structures	PPT	DS_Presentation.pptx	19 May 2025	Uploaded	View
Web Development	PBL	Web_Project.zip	18 May 2025	Uploaded	View
Database Management Systems	Assignment	DBMS_Notes.docx	17 May 2025	Uploaded	View

Figure 5.5: My Assignment Page

Student Assignment Management System

Dashboard
Subjects
View Submissions
Marks Entry
Reports
Logout

Teacher

Teacher Dashboard

Welcome back, Teacher!

5
Subjects

45
Total Students

28
Eval assignments

32
Evaluated

Recent Submissions

[View All](#)

Student Name	Subject	Type	File Name	Date
Ravi Kumar	Java Programming	Assignment	Java_Notes.pdf	20 May 2025
Sita Devi	Data Structures	PPT	DS_Presentation.pptx	19 May 2025
Amit Singh	Web Development	PBL	Web_Project.zip	18 May 2025

Figure 5.6: Teacher Dashboard

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CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

CONCLUSION:

The **Student Assignment Management System** successfully addresses the challenges involved in traditional assignment submission and management processes. The system provides a centralized platform where students can upload assignments, PPTs, and PBL files while teachers can review submissions, assign marks, and download reports efficiently.

The project integrates student login functionality, subject management, assignment uploads, teacher dashboards, marks entry, file viewing, Excel report generation, and database handling using Spring Boot and MySQL. This makes the application suitable for educational institutions for managing student submissions digitally.

The assignment handling mechanism is one of the major outcomes of the project. It stores uploaded files securely, organizes submissions according to subjects, and allows teachers to access student records in a structured manner. The system improves transparency, reduces paperwork, and minimizes manual errors during assignment evaluation.

The project also creates a strong foundation for future improvements such as cloud storage integration, notification systems, plagiarism detection, and mobile application support.

FUTURE ENHANCEMENTS:

- Integrate email and SMS notifications for assignment reminders and submission alerts.
- Develop Android and iOS mobile applications for easier access.
- Add plagiarism detection functionality for assignment verification.
- Introduce cloud-based storage for improved file management and scalability.
- Provide analytics dashboards for student performance and submission statistics.
- Add support for assignment deadlines and automatic late submission tracking.
- Implement role-based access with enhanced security features.
- Integrate AI-based feedback and performance recommendations.
- Improve security using activity logging and advanced authentication mechanisms.

REFERENCES

1. Java Official Documentation – <https://www.java.com/>
2. Spring Boot Documentation – <https://spring.io/projects/spring-boot>
3. MySQL Official Documentation – <https://dev.mysql.com/doc/>
4. HTML Documentation – <https://developer.mozilla.org/en-US/docs/Web/HTML>
5. CSS Documentation – <https://developer.mozilla.org/en-US/docs/Web/CSS>
6. JavaScript Documentation – <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
7. Spring Data JPA Documentation – <https://spring.io/projects/spring-data-jpa>
8. Apache POI Documentation – <https://poi.apache.org/>
9. Software Engineering: A Practitioner's Approach — Roger S. Pressman
10. Database System Concepts — Abraham Silberschatz, Henry F. Korth, S. Sudarshan

APPENDIX AND SAMPLE CODE

A.1 DATASET / USER DATA DESCRIPTION

The database stores student information, teacher details, subjects, assignment uploads, marks records, uploaded file paths, and evaluation details. The system maintains subject-wise submissions and allows teachers to review uploaded documents efficiently.

Table	Purpose
students	Stores student profile, roll number, section, branch, and login details.
subjects	Stores available subject names for assignment submission.
marks	Stores assignment records, uploaded file paths, subject details, and marks.
teachers	Stores teacher profile and login details.
assignment_files	Stores uploaded assignment document information.
ppt_files	Stores uploaded PPT file details.
pbl_files	Stores uploaded PBL document details.
reports	Stores generated Excel reports and marks records.
uploads	Stores file upload paths and submission history.
notifications	Stores assignment alerts and updates.

A.2 SYSTEM MODULES / ARCHITECTURE

1. Authentication Module

Student login verifies users using roll number and password details.

2. Subject Module

Displays available subjects for assignment uploads.

3. Assignment Upload Module

Allows students to upload Assignment, PPT, and PBL files.

4. File Management Module

Maintains uploaded file storage and retrieval.

5. Teacher Dashboard Module

Displays uploaded student files and marks management options.

6. Marks Module

Allows teachers to enter and update marks.

7. Report Module

Supports Excel report generation and download.

8. Database Module

Stores student records, submissions, subjects, and marks.

A.3 TRAINING AND EXECUTION SETUP

Execution Setup

- Install Maven dependencies using project build tools.
- Configure MySQL credentials in application.properties file.
- Run the Spring Boot application.
- Open the application in browser using localhost URL.
- Login using student or teacher credentials.

Execution Flow

Start Application → Connect Database → Student Login → Select Subject → Upload Files → Store Data → Teacher Dashboard → View Files → Assign Marks → Download Excel Report

A.4 DEPLOYMENT DETAILS

The application can be deployed on a Spring Boot compatible hosting environment with MySQL database support. Environment configurations should be secured properly for database credentials and application settings.

Deployment Features

- Server-side assignment management system

- Persistent MySQL database storage
- Student file upload support
- Subject-wise assignment handling
- Excel report generation
- Teacher dashboard functionality
- Secure role-based access

A.5 RESULTS

The completed project demonstrates successful student login, subject-wise assignment upload, teacher file viewing, marks entry, database storage, and Excel report generation.

A.6 BENEFITS OF THE SYSTEM

- Reduces manual assignment collection work.
- Provides centralized file management.
- Supports subject-wise assignment tracking.
- Allows teachers to evaluate submissions easily.
- Generates reports and marks efficiently.
- Maintains structured student submission records.
- Improves transparency and assignment management process.

A.7 SAMPLE CODE

Student Login Model (Student.java)

```
package com.teamcomet.backend.model;

import jakarta.persistence.*;

@Entity
@Table(name="students")

public class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String rollNo;

    private String name;

    private String section;

    private String branch;

    private String dob;

    public Long getId() { return id; }

    public String getRollNo() { return rollNo; }

    public void setRollNo(String rollNo){
        this.rollNo = rollNo;
    }
}
```

Subject Controller

```
package com.teamcomet.backend.controllers;

import org.springframework.web.bind.annotation.*;

import java.util.Arrays;

import java.util.List;

@RestController
@RequestMapping("/api/subjects")
@CrossOrigin(origins="*")
```



```

public class SubjectController {

    @GetMapping
    public List<String> getSubjects() {
        return Arrays.asList(
            "Data Structures",
            "COA",
            "COSM",
            "DE",
            "JAVA",
            "R Programming"
        );
    }
}

```

Marks Controller

```

@RestController
@RequestMapping("/api/marks")
public class MarksController {

    @Autowired
    MarksRepository repo;

    @GetMapping("/subject/{subject}")
    public List<Marks> getBySubject(
        @PathVariable String subject){
        return repo.findBySubject(subject);
    }

    @PutMapping("/update/{id}")
    public String updateMarks(
        @PathVariable Long id,
        @RequestParam int marks){
        Marks m=repo.findById(id).orElse(null);

```

```

    if(m==null)

        return "Not Found";

    m.setTotalMarks(marks);

    repo.save(m);

    return "Marks Updated";

}

}

```

Student File Upload API

```

@PostMapping("/upload")

public String upload(

    @RequestParam String subject,

    @RequestParam String teamId,

    @RequestParam String studentName,

    @RequestParam MultipartFile file){

    String fileName=file.getOriginalFilename();

    return "File Uploaded Successfully";

}

```

Teacher Dashboard JavaScript

```

async function save(id){

    let val =

    document.getElementById(

    `t-${id}`).value;

    await fetch(

    `/api/marks/update/${id}?marks=${val}`,

    {method:'PUT'}

    );

    alert("Marks Updated"); }

```

Database Query

```
SELECT * FROM marks
```

```
WHERE subject='JAVA';
```

```
UPDATE marks
```

```
SET total_marks=90
```

```
WHERE id=1;
```